COURSE NAME: PLANT PROTECTION

COURSE CODE: NR 430

- PATHOLOGY-
LECTURE 1:

1. INTRODUCTION

Plants cover the most area of the earth's living environment as trees, grasses, flowers, and so on. Plants play different important roles in the environment such as ecosystem balance and food supplement for animals and humans.

The study of plant pathogens belongs to the branch of biology known as **plant pathology**.

Diseases in plants are classified according to the two types of origins and these are **biotic and/or abiotic** origin. *Biotic (infectious)* diseases are developed owing to microbial infection, while abiotic (noninfectious) diseases are developed due to unfavorable environmental conditions, such as lack or excess of nutrients, moisture, light, and the presence of toxic chemicals in air or soil etc.

Plant pathology therefore focuses on the study of the organisms and the environmental factors that cause disease in plants; the mechanisms or methods by which these factors induce disease in plants and the methods of preventing or controlling disease aiming at reducing the damage it causes.

Essentially, plant pathology is correlated with other sciences such as entomology, bacteriology, mycology, virology, and weed science due to the visible effects of insects, bacteria, fungi, viruses/viroids, and weeds on plants, respectively

Terms and concepts in plant pathology:

- **Pathogen**: is defined as an organism causing disease to its host
- **Phytopathogens**: an organism which causes diseases in plants
- **Biotic diseases**: these are diseases caused by the pathogen (fungi, bacteria or viruses)
- **Abiotic diseases**: these are diseases which are caused by non-living agents like weather, soils, chemicals, or mechanical injuries. They are also referred to as non-infectious diseases.

- **parasitism:** an organism (parasite) lives on or in another organism (host) to obtain its required nutrition
- **Epiphytes**: any plant that grows upon another plant or object merely for physical support. Epiphytes have no attachment to the ground or other obvious nutrient source and are not parasitic on the supporting plants
- **Saprophytes:** An organism, especially a fungus or bacterium that lives on and gets its nourishment from dead organisms or decaying organic material. Saprophytes recycle organic material in the soil, breaking it down into in simpler compounds that can be taken up by other organisms.
- **pathogenicity**: this is the ability of an organism to cause a disease
- **Virulence:** is the disease producing power of an organism or ability of a pathogen or microbe to infect or damage a host. It can also be defined as the level of severity of the disease symptom.
- **Infectious diseases**: these are diseases that result from infection of a plant by a pathogen
- **Acquired resistance**: Plant resistance to disease activated after inoculation of the plant with certain microorganisms or treatment with certain chemical compounds.
- **Parasite**: An organism living upon or in another living organism (the host) and obtaining the food from the invading host.
- . **Biotroph**: A plant pathogenic fungus that requires living host cells i.e. an obligate parasite.
- **Hemibiotroph**: A plant pathogenic fungus that initially requires living host cells but after killing the host cell grows on the dead and dying cells.
- **Necrotroph**: A pathogenic fungus that kills the host and survives on the dying and dead cells.
- **Pathogenesis**: It is a process caused by an infectious agent (pathogen) when it comes in contact with a susceptible host.
- **Infection**: The initiation and establishment of a parasite within a host plant.
- . **Inoculum**: That portion of pathogen which is transferred to plant and cause disease.

- **Invasion**: The penetration and spread of a pathogen in the host.
- **Colonization**: The growth of a pathogen, particularly a fungus, in the host after infection is called colonization.
- **Inoculum potential**: The growth or threshold of fungus available for colonization at substratum (host).
- **Symptoms**: The external and internal reaction or alterations of a plant as a result of disease.
- **Incubation period**: The period of time between penetration of a pathogen to the host and the first appearance of symptoms on the plant.
- **Disease cycle**: The chain of events involved in disease development.
- **Disease syndrome**: The set of varying symptoms characterizing a disease are collectively called a syndrome.
- **Alternate host**: Plants not related to the main host of parasitic fungus, where it produces its different stages to complete one cycle (heteroecious).
- **Collateral host**: The wild host of same families of a pathogen is called as collateral host.
- **Predisposition**: The effect of one or more environmental factors which makes a plant vulnerable to attack by a pathogen.
- **Physiologic race**: One or a group of microorganisms similar in morphology but dissimilar in certain cultural, physiological or pathological characters.
- **Symbiosis**: A mutually beneficial association of two or more different kinds of organisms.
- **Mutualism**: Symbiosis of two organisms that are mutually helpful or that mutually support one another.
- **Antagonism**: The counteraction between organisms or groups of organisms.
- **Mutation**: An abrupt appearance of a new characteristic in an individual as a result of an accidental change in genes present in chromosomes.
- **Disease**: Any deviation in the general health, or physiology or function of plant or plant parts, is recognized as a disease.

- **Crop Damage**: It is defined as any reduction in the quality or quantity of yield or loss of revenue resulting from crop injury.
- **Deficiency:** Abnormality or disease caused by the lack or abnormal level of availability of one or more essential nutrient elements.

2. THE CONCEPT OF DISEASE IN PLANTS

Unlike human beings and animals, it is not known whether plants also feel pain or discomfort and plants do not speak or otherwise communicate with us, it is difficult to pinpoint exactly when a plant is diseased especially if there are no visible signs.

A plant is considered to be healthy or normal, if it's able to carry out its physiological functions to the best of its genetic potential.

i. The meristematic (cambium) cells of a healthy plant should **divide and differentiate**,
and different types of specialized cells (like the phloem)must be able to **absorb water and nutrients from the soil**; translocate these to all plant parts;

ii. a plant must be able to carry out **photosynthesis**, translocate, metabolize, or store the photosynthetic products;

iii. **and must be able to produce seed** or other reproductive organs for survival and multiplication.

Therefore, if the plant is unable to carry out most of these activities due to the presence of the pathogen or unfavorable environmental condition, then it is said to be **diseased**.

Plant disease, then, can be defined as the series of invisible and visible responses of plant cells and tissues to a pathogenic organism or environmental factor that result in adverse changes in the function, or integrity of the plant and may lead to partial impairment or death of plant parts or of the entire plant. The kinds of cells and tissues that become affected determine the type of physiological function that will be disrupted first. For example, infection of roots may cause roots to rot and make them unable to absorb water and nutrients from the soil.

The first step of plant disease treatment is observation of definite and clear symptoms on the plants. These symptoms give an initial indication for the type and cause of plant disease, which may end with the death

The infected part of the plant gives an indication of the type of plant disease, such as infected root which is usually correlated with root-rot disease.

Plant diseases can be classified and identified according to **several parameters** and some of them are:

- **i.** disease symptoms,
- **ii.** infected organ,
- **iii.** infected plant type,
- **iv.** and the type of phytopathogen.

Identifying the phytopathogen is considered to be more useful criterion used for plant disease classification, because it easily determines the disease cause, potential disease complications, and offer possible control methods.

Pathogenic microorganisms cause diseases in plants by;

- **i.** Disturbing the metabolism of plant cells through enzymes, toxins, growth regulators, and other substances they secrete and by absorbing foodstuffs from the host cells for their own use.
- **ii.** Some pathogens may also cause disease by growing and multiplying in the xylem or phloem vessels of plants, thereby blocking the upward transportation of water or the downward movement of sugars, respectively, through these tissues

Thousands of plant diseases exists with some affecting a lot of plant species while just a few are specific. It is important to note that a single plant species can also be affected by a hundred or more diseases.

LECTURE 2.

1. TYPES OF PLANT DISEASES AND BASIC PROCEDURES IN THE DIAGNOSIS OF PLANT DISEASES

The plant disease diagnosis depends on the exact determination of the disease cause. Generally, there are two plant disease causes: the **pathogens** and/or **environmental factors**. The former leads to **infectious** diseases, while the latter leads to **non-infectious** diseases.

The most common method of diagnosing a plant disease is by grouping them according to the symptoms they cause

i. do they cause root rots, wilts, leaf spots, blights, rusts, smuts,
ii. the plant organ they affect (root diseases, stem diseases, foliage diseases),
iii. orto the types of plants affected (field crop diseases, vegetable diseases, tree diseases, turf diseases etc.

1.1. Infectious diseases

There are wide range of phytopathogens which cause infectious plant diseases such as fungi, bacteria, viruses, viroids, and protozoa. The infectious disease means the ability of phytopathogen to transfer diseases from the infected plant to another healthy one and causes the same disease and the same symptoms. Most phytopathogens can inhabit the internal environment of plants; however, some others can live on the plant surface such as some fungi, bacteria, and parasitic higher plants

Infectious or biotic plant diseases

- Diseases caused by fungi
- Diseases caused by bacteria
- Diseases caused by parasitic higher plants and green algae
- Diseases caused by viruses and viroids
- Diseases caused by nematodes

Noninfectious, or abiotic plant diseases

- Diseases caused by too low or too high temperatures
- Diseases caused by lack or excess of soilmoisture
- Diseases caused by lack or excess of light
- Diseases caused by lack of oxygen
- Diseases caused by air pollution

- Diseases caused by nutrient deficiencies
- Diseases caused by mineral toxicities
- Diseases caused by soil acidity or alkalinity (soil pH)

1.1.1. Diseases caused by parasitic higher plants

Some plant diseases are developed due to growing of certain plants attached on or in other plants, where they take all required nutrients without benefit sharing; these plants are called parasitic higher plants. This abnormal relationship leads to weakness of healthy or host plant. The parasitic higher plants are usually found attached with the surface of the host plant, such as dodder, witchweed, and broomrape.

1.1.2. Diseases caused by nematodes

The nematodes are one of most common phytopathogens which have definite symptoms. These symptoms only appear in the infected site. The nematode infections in or on plants are widely distributed especially in proper environments such as moderate temperature and high humidity.

2.1.3. Diseases caused by fungi

Interestingly, there are two main types of fungi appearing on plants: pathogenic and saprophytic. The pathogenic fungi live in or on plant tissues and cause serious complications for the vital physiological functions of plants, while saprophytic ones live in or on dead tissues. Accordingly, the diagnosis of plant disease must be exactly carried out. The exact diagnosis and determination of fungi take place by microscopical examination to identify the mycelial morphological characteristics, whatever fruiting structures and spores. After complete identification for the fungus and the symptoms of plant disease, the latter should be compared with that reported in the reference. This study will exactly determine whether the fungus is a pathogen or a saprophyte. Although microscopical examination is an essential and effective method for fungal identification, it only sometimes cannot lead to exact identification due to the absence of fungal fruiting structures and spores on infected plant tissue.

Therefore, an alternative method must be used, such as using selective media for isolation, identification, or promotion of sporulation. On the other hand, some fungi need to be incubated under certain temperature, aeration, or light conditions to produce spores

2.1.4. Diseases caused by bacteria and mollicutes

The appearance of bacterial growth in or on plant tissues means that bacterial plant disease may be present, because saprophytes may be present. Therefore, accurate bacterial identification must be carried out by using microscopical examination and physiological parameter determination.

The selective media are essentially used in the bacterial identification to determine the bacterial genus and species in some cases. Moreover, the confirmatory test of bacterial pathogenicity may be carried out by inoculation of single pure bacterial colony in the healthy plant, reproducing the same symptoms that appeared on the infected one. Moreover, immunodiagnostic techniques or serodiagnostic assays can be used, such as agglutination and precipitation, fluorescent antibody staining, and enzyme-linked immunosorbent assay (ELISA).

There are several advantages for these techniques such as quite sensitivity, fairly specificity, rapid, and easy to perform.

The molecular biological techniques are also widely used.

Mollicutes ;

There are uncommon microorganisms called **mollicutes and** similar to bacteria. These microorganisms are very small where they must be examined by an electron microscope. Mollicutes have polymorphism(2 or more variants of a particular DNA sequence) and lack cell wall–like mycoplasma. These microorganisms habit the young phloem cells as a convenient host, and cause severe plant diseases such as plant stunting, yellowing or reddening of leaves, proliferation of shoots and roots, production of abnormal flowers, and eventual decline and death of the plant. Mollicutes cannot be cultured on nutrient media except for the genus *Spiroplasma*. Mollicutes can be diagnosed by several parameters, such as symptoms determination, grafting, transformation, microscopical examination, susceptibility to tetracyclines, and so on.

2.1.5. Diseases caused by viruses and viroids

There are distinctive types of plant diseases caused by viruses/viroids. These diseases have definite and clear symptoms, which easily support disease diagnosis and are considered main advantage. Apart from this advantage, some recent techniques are widely used for disease diagnosis and virus identification, such as virus transmission tests to specific host plants by sap inoculation, grafting, certain insect, nematode, fungus, and mite vectors.

Moreover, serodiagnostic tests are used for this purpose such as enzyme-linked immunosorbent assays (ELISA), gel diffusion tests, micro-precipitin tests, and fluorescent antibody staining. The electron microscopy techniques as negative staining of virus particles in leaf dip or purified preparations are also used, as well as immune-specific electron microscopy.

On the other hand, there are more accurate techniques used for disease diagnosis and virus/viroid identification, such as electrophoretic tests and hybridization of commercially

available radioactive DNA complementary to a certain virus DNA or RNA, or viroid RNA, with the DNA or RNA present in plant sap and attached to a membrane filter (immunoblot).

2.1.6. Diseases caused by more than one pathogen

Sometimes, some plants are exposed to co-infection by two or more pathogens, which lead to the same or different disease symptoms. Therefore, the differentiation and identification of these pathogens are very essential to exactly determine the disease cause. The differentiation and subsequently identification are carried out by all techniques that are mentioned above.

2. PARASITISM AND PATHOGENICITY

The term parasitism called on the state in which an organism (parasite) lives on or in another one (host) to obtain its required nutrition. Usually, parasitism is correlated with **pathogenicity**, which means the ability of an organism to cause a disease.

Parasitism in some cases leads to a benefit relationship called **symbiosis**, in which both plant and organism alternate the benefits, such as bacterial nodules in the roots of legume plants and the mycorrhizal infection of feeder roots of most flowering plants. In the case of parasitism-pathogenicity relationship, the plant is diseased with the appearance of different symptoms such as increased respiration, disintegration or collapse of cells, wilting, abscission, abnormal cell division and enlargement, and degeneration of specific components such as chlorophyll.

The most common plant pathogens are fungi, bacteria, mollicutes, parasitic higher plants, parasitic green algae, nematodes, protozoa, viruses, and viroids. These parasites cause serious plant diseases, because they have the ability to penetrate the plant tissues to feed and proliferate in it, and withstand the conditions in which the host lives. These pathogens are also called **obligate parasites** because they can only live in their living hosts.

On the other hand, there are certain pathogens such as most fungi and bacteria can live on either living or dead hosts and on various nutrient media, so they are called **nonobligatory parasites**. Some nonobligatory parasites can grow saprophytically on dead organic matter, and therefore called semi-biotrophs/**facultative saprophytes**. There is a type of a life called facultative parasitism, in which an organism grows saprophytically (necrotrophs); however, under certain conditions, they attack living plants and cause a disease; these parasites are called **facultative parasites**. The type or degree of parasitism does not affect the disease severity. For instance, many diseases caused by weakly parasitic pathogens are much more damaging to a plant than others caused even by obligate parasites.

LECTURE 3

1.0 COMMON PLANT DISEASES AND THEIR SYMPTOMS

1.1 FUNGAL DISEASES

1.1.1 Introduction

Fungi are eukaryotic organisms; i.e., their cells contain membrane-bound organelles and clearly defined nuclei. They are small, generally microscopic and usually filamentous, branched, spore-bearing organ- isms that lack chlorophyll. Fungi have cell walls that contain chitin and glucans (but no cellulose) as the skeletal components.

Some fungi, known as **obligate parasites or biotrophs**, can grow and multiply only by remaining, in association with their host plants in their entire life. Others, known as **non-obligate parasites,** require a host plant for part of their life cycles but can complete their cycles on dead organic matter, or they can grow and multiply on dead organic matter as well as on living plants.

1.1.2 Characteristics of plant pathogenic fungi

i. Morphology

Most fungi have a filamentous vegetative body called a **mycelium.** The mycelium branches out in all directions. The individual branches of the mycelium are **called hyphae.**

ii. Reproduction

Fungi reproduce chiefly by means of spores. Spores are reproductive bodies consisting of one or a few cells and may be formed asexually,or as the result of sexual fertilization.

In some fungi, **asexual spores** are produced inside a sac called a ***sporangium***. Some of these spores can swim by means of flagella and are called ***zoospores***. Other fungi produce asexual spores called **conidia** by the cutting off of terminal or lateral cells from special hyphae called **conidiophores.**

Sexual reproduction occurs in most groups of fungi. In Zygomycetes, two cells (gametes) of similar size and appearance unite and produce a zygote called a **zygospore.**

iii. Dissemination

Zoospores are the only fungus structures that can move by themselves. Zoospores, however, can move for only very short distances like few millimeters or centimeters in rare instances. Only **myxomycetes, oomycetes**, and **chytridiomycetes** produce zoospores. The great majority of plant pathogenic fungi depend for their spread from plant to plant and to different parts of the same plant on hyphal growth or chance distribution by such agents as wind, water, birds, insects, other animals, and humans. Fungi are disseminated primarily in the form of spores.

1.1.3 Signs and symptoms caused by fungi on plants

In most cases, Fungi cause local or general symptoms on their hosts. As such symptoms may occur separately or concurrently or may follow one another.

In general, fungi cause local or general **necrosis** of plant tissues, and they often cause reduced growth **(stunted growth or hypotrophy)** of plant organs or entire plants. A few fungi cause excessive growth of infected plants or plant parts (***hypertrophy or hyperplasia***) .

Symptoms associated with necrosis include *leaf spots, blight, scab, rots, damping-off, dieback, and canker*. Symptoms associated with hyperplasia include galls, warts, and leaf curls.

In some instances, the fungus infecting the plant may produce growth or structures on the plant, stems, or leaves such as masses of mycelium or aggregates of spores with a characteristic appearance. These developments are referred to as **signs of infection**, in contrast to symptoms, which refer specifically to **the plant or plant tissue.**

The most commonnecrotic symptoms of fungal infection are as follows:

i. **Leaf spots**: Localized lesions on host leaves consisting of dead and collapsed cells
ii. **Blight**: this causes general and extremely rapid browning and death of leaves, branches, twigs, and floral organs.
iii. **Canker**: Localized necrotic lesion on a stem or fleshy organ, often sunken, of a plant
iv. **Dieback**: Extensive necrosis of twigs beginning at their tips and advancing toward their bases
v. **Root rot:** Disintegration or decay of part or all of the root system of a plant

vi. **Scab:** Localized lesions on host fruit, leaves, tubers, etc., usually slightly raised or sunken and cracked, giving a scabby appearance

vii. **Soft rots and dry rots**: Maceration and disintegration of fruits, roots, bulbs, tubers, and fleshy leaves

viii. **Damping-off**: Rapid death and collapse of very young seedlings

ix. **Basal stem rot**: Disintegration of the lower part of the stem

x. **Galls**: Enlarged portions of plant organs (stems, leaves, blossoms, roots)

xi. **Warts**: Wart-like protuberances on tubers and stems

xii. **Witches'-brooms**: Profuse, upward branching of twigs

xiii. **Leaf curls**: Distortion, thickening, and curling of leaves

xiv. **Wilt:** Generalized loss of turgidity and drooping of leaves or shoots

xv. **Rust:** Many small lesions on leaves or stems, usually of a rusty color

xvi. **Smut:** Seed or a gall filled with the mycelium or black spores of the smut fungi

xvii. **Mildew**: Areas on leaves, stems, blossoms, and fruits, covered with whitish mycelium

In many diseases, the fungal pathogen grows, or produces various structures on the surface of the host. These structures may include mycelium, fruiting bodies, and spores, and are called signs. Signs refer only to the appearance of the infected plants or plant tissues.

For example in mildews, one sees mostly the signs consisting of a whitish, downy or powdery growth of fungus mycelium and spores on the plant leaves, fruit, or stem, whereas the symptoms consist of chlorotic or necrotic lesions on leaves, fruit, and stem, reduced growth of the plant, and so on.

To be sure that the disease is fungal, part of the infected parts e.g leaf or fruit can be isolated and put on the petri-dish for laboratory analysis using scientific methods such as **Polymerise Chain Reaction (PCR).**



Witches broom





Late blight in tomatoes



Fruit cankers





Powdery mildew on tomato **Damping off**

1.2 BACTERIAL DISEASES

1.2.1 Introduction

Bacteria are generally single-celled microorganisms whose genetic material (DNA) is not bound by a membrane and therefore is not organized into a nucleus.

Most bacteria are strictly saprophytic and as such are beneficial to humans because they help decompose the enormous quantities of organic matter produced yearly by humans, animals, and factories as waste products or by the death of plants and animals. Several species cause diseases in humans, such as tuberculosis, pneumonia, and typhoid fever, and a similar number cause diseases in animals, such as brucellosis and anthrax.

Bacteria are in different shapes, some are rod shaped, spherical, spiral, or filamentous (threadlike). Some bacteria can move through liquid media by means of ***flagella***, whereas others have no flagella and cannot move themselves

1.2.2 Characteristics of plant pathogenic bacteria

i. Morphology

Majority of the plant pathogenic bacteria are rod-shaped.

The cell walls of bacteria of most species are enveloped by a viscous layer forming a definitive mass around the cell, is called a ***capsule.*** All the material inside the cell wall constitutes the protoplast.

Rod-shaped phyto-pathogenic bacteria reproduce by the asexual process known as ***binary fission, or fission.***

This occurs by the inward growth of the cytoplasmic membrane toward the center of the cell, forming a transverse membranous partition dividing the cytoplasm into two approximately equal parts. The two layers of cell wall material, are then synthesized between the two layers of the membrane.

When the formation of these cell walls is completed, the two layers separate, splitting the two cells apart.

Bacteria reproduce at an **astonishingly rapid rate**. Under favorable conditions, bacteria may divide every 30 to 50 minutes, one bacterium becoming 2, 2 becoming 4, 4 becoming 8, 8 become 16 and so on. Over a million bacteria can be produced in a day.

ii. Transmission and infection

In order for a bacterium to produce a disease in a plant, the bacterium must first invade the plant tissue and multiply. Bacterial pathogens enter plants through wounds, principally produced by adverse weather conditions, humans, tools and machinery, insects, and nematodes, or through natural openings such as stomata, lenticels (pores in the stem for gaseous exchange), hydathodes, nectar-producing glands, and leaf scars.

Most foliage invaders are spread from plant to plant by windblown rain or dust. Humans disseminate bacteria through *cultivation, grafting, pruning, and transporting diseased plant material*. Animals, including insects and mites, are other common transmission agents

When conditions are unfavorable for growth and multiplication, bacteria remain dormant on or inside plant tissue. Some, such as the **crown gall bacterium**, may survive for months or years in the soil.

Bacterial diseases are influenced greatly by **temperature and moisture**. Often, a difference of only a few degrees in temperature determines whether a bacterial disease will develop. In most cases, moisture as a water film on plant surfaces is essential for establishing an infection.

iii. Ecology and spread

Almost all plant pathogenic bacteria develop mostly in the host plant as parasites, on the plant surface, especially buds, as **epiphytes**, and partly in plant debris or in the soil as **saprophytes.**

The main classes of bacteria that causes diseases in plants are **agrobacteria** which are rod-shaped and possess flagella, **clavibacter** which are mainly non-motile, ***erwinia*** which are straight rods and are motile by means of flagella and the last group is **pseudomonas** which are straight or curve shaped and are most abundant in soils, fresh water and marine environments. Other are ralstonia, xylella and xanthomonas groups.

1.3 Symptoms and signs caused by bacterial disease

Bacterial diseases can be grouped into **four** broad categories based on the extent of damage to plant tissue and the symptoms that they cause, which may include:

(i) Vascular wilt,

Vascular wilt results from the bacterial invasion of the plant's vascular system. The subsequent multiplication and blockage prevents movement (translocation) of water and nutrients through the xylem of the host plant. Drooping, wilting, or death of the aerial plant structure may occur; examples include bacterial wilt of sweet corn, tobacco, tomato, and cucurbits (e.g., squash, pumpkin, and cucumber).

(ii) Necrosis

Pathogens can cause necrosis by secreting a toxin (poison). Symptoms include formation of leaf spots, stem blights, or cankers.

(iii) Soft rot

Soft rot diseases are caused by pathogens that secrete enzymes capable of decomposing cell wall structures, thereby destroying the texture of plant tissue—i.e., the plant tissue becomes macerated (soft and watery). Soft rots commonly occur on fleshy vegetables such as potato, carrot, eggplant, squash, and tomato.

(iv)Tumors.

Tumor diseases are caused by bacteria that stimulate uncontrolled multiplication of plant cells, resulting in the formation of abnormally large structures.

1.3Examples of some specific bacterial diseases

Below is a list of examples of some common bacterial diseases.

i. Bacterial spots and blight

These appearas spots of various sizes on leaves, stems, blossoms, and fruits. In some bacterial diseases the spots continue to advance rapidly and the diseases are then called blights.

In severe infections the spots may be so numerous that they destroy most of the plant surface and the plant appears blighted, thus producing large areas of dead plant tissue and blighted plants.

In **dicotyledonous plants** the bacterial spots on some hosts are restricted by large veins, and the spots appear angular. For the same reason, bacterial spots on monocotyledonous plants appear as streaks or stripes.



Bacterial leaf spot on pepper



Bacterial spot (xanthomonas)



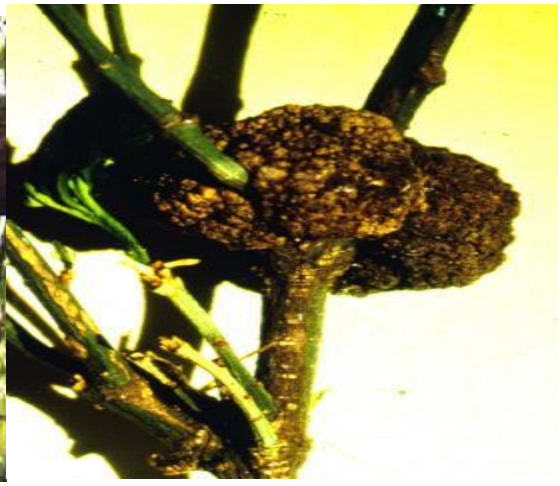
Tomato bacterial blight



Bacterial blight in beans



Galls on fruit trees



Crown Gall on fruit tree (*Agrobacterium tumefaciens***)**

1.4 Management and control of plant diseases

Bacterial diseases of plants are usually very difficult to control. Frequently, a combination of control measures is required to combat a given bacterial disease. The following measures can help to control bacterial as well as other plant diseases.

- **Chemical controls**: involve the use of chemicals like fungicides, bactericides to control plant disease. Chemical controls can be effective, but it is important to properly identify the problem which may require submitting samples to a diagnostic lab for testing.
- **Genetic controls**: involve breeding disease resistant plants which creates new cultivars or varieties that perform better.
- **Cultural practices**: this involves the use of various practices that aim at avoiding the provision of conducive environment for the survival and multiplication of the pathogen. For example, crop rotation to disrupt year-to-year pest cycles, sanitizing tools after using them on diseased plants to prevent further spread, planting warm seasoned crops after the soil has warmed to prevent root rot, watering plants in the morning so they can dry during the day and remain dry through the night to prevent

fungal, and bacterial diseases, and pruning and trimming plants in such a way that it promotes airflow and more light penetration to prevent infection.

- **Regulatory controls:**these are based on exclusion of pathogenic organisms, highly susceptible host plants, and alternate host plants from a particular area.
- **Biological controls**: this essentially involve using nature to prevent or treat diseases
- **Sanitation practices:** aiming at reducing the inoculum in a field by removing and burning infected plants or branches, and at reducing the spread of bacteria from plant to plant. This includes cleaning of the farm equipment and proper weeding.
- **Adjusting fertilizing and watering**: this is done so that the plants are not extremely succulent during the period of infection may also reduce the incidence of disease
- **Crop rotation**: this can be very effective with bacteria that have a limited host range, but is impractical and ineffective with bacteria that can attack many types of crop plants
- **Seed treatment** with hot water at about 50 °C (120 °F) is also effective against management of bacterial diseases in carrots, eggplant, pepper and tomatoes.
- **destruction** of all plant materials that may harbor pathogenic fungi
- **Biosecurity measure** are also important to avoid the transfer of disease pathogens from an infected area to another.
- **Isolation** or, as it is usually called, purification of viruses is a very effective method of managing viral pathogens.